

A 13.56MHz RFID Device and Software for Mobile Systems

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Abstract— We report a new electronic device and software with the ability to write data to and read data from contact-less mobile systems adapting electromagnetic inductive coupling. Establishing electromagnetic-field simulation based on the Finite Element Method (FEM) and using a new magnetic material, we have developed a small, thin Radio Frequency Identification (RFID) device on metallic housing. Furthermore, device-driver and network-library software have been developed for use in cashless mobile systems as electronic money. (*Abstract*)

Keywords—component; RFID; mobile systems; metallic housing; electromagnetic-field simulation; FEM; eddy current; magnetic material; device driver software

I. INTRODUCTION

RFID systems using a contact-less IC card or IC tag are being used as automatic ticket checkers used in the railway stations, security systems checking people going entering or exiting from buildings, and electronic-money systems. New fabrication techniques and algorithms providing better security have been developed so mobile systems using RFID communication systems are now possible.

13.56MHz-RFID systems have realized near-field communication by adopting electromagnetic induction. An electromagnetic field radiated from a loop antenna provided in the reader/writer is coupled by electromagnetic induction to a loop antenna in the IC card. However, if the loop antenna is installed on a metallic housing such as that of a PDA (personal digital assistant) or a mobile phone, the loop antenna for the reader/writer cannot efficiently radiate an electromagnetic field to the IC card: due to eddy current loss, the communication range between the IC card and reader/writer is narrow. Current non-portable products apply magnetic sheets to minimize the influence of metallic housing, however a small, thin magnetic sheet for mobile systems is not effective in allowing communication with a reader/writer at some distance.

A 13.56MHz-RFID device is usually designed using the electromagnetic field simulation and the circuit simulation. [1] However, current moment methods are not particularly useful for designing a device with a magnetic sheet. [2] To solve this problem, we developed a new method of electromagnetic-field simulation based on the finite element method (FEM), which allows us to quickly determine the relationship between the

magnetic sheet and a loop antenna in order to minimize the influence of eddy current loss.

We have also developed two new software programs, which can be used to construct the architecture for driver platforms for E-commerce applications. The device driver controls reader/writer devices connected to IC cards or other devices and operates cashless mobile systems. Network-library software ensures secure identification of the user and secure transmission between server and client.

II. FEM ELECTROMAGNETIC-FIELD SIMULATION

We can calculate the performance of RFID tags and reader/writers using FEM electromagnetic-field simulation. The performance of an RFID device is simulated using the following steps.

Step 1. Simulating the inductance and resistance of the device.

Step 2. Simulating the mutual inductance and capacitance between the tag and the reader/writer.

Step 3. Calculating the response of the tag to the waveform input to the reader/writer using a circuit simulator.

A. Simulating inductance and resistance

We have developed a method to precisely calculate the resistance of an RFID device having a magnetic sheet using FEM electromagnetic field simulation. We checked the accuracy of our method by comparing the value derived using the simulation and the experimental value. We checked the differences in eight types of devices. Figs. 1 and 2 compare the simulations and the experimental results. The differences are less than 10% as regards both resistance and inductance.

B. Simulating mutual inductance and capacitance

We can calculate the mutual inductance and capacitance between the device and the reader/writer. In this case we used three-dimensional models of the device and the reader/writer. Using three-dimensional models enables to change the relative position between the device and the reader/writer. Fig. 3 shows a model of the device and the reader/writer.

C. *Simulating the response of the device using a circuit simulator*

We can calculate the response of the device to the waveform input to the reader/writer using a circuit simulator. The parameters inputted the circuit simulator are the inductance, the resistance, the mutual inductance, and the mutual capacitance. The mutual inductance varies with the distance between the device and the reader/writer and using our algorithm we can calculate the range over which the device and the reader/writer can communicate.

D. *Calculating the degree of influence of metallic housing*

We can calculate the effect of the magnetic sheet on the metallic housing using simulation. Fig. 4 shows the effect of a magnetic sheet by comparing the value of the device with a magnetic sheet and the value of the device without a magnetic sheet. We can see that the device having a magnetic sheet is more stable and can obtain higher coupling coefficient on metallic housing.

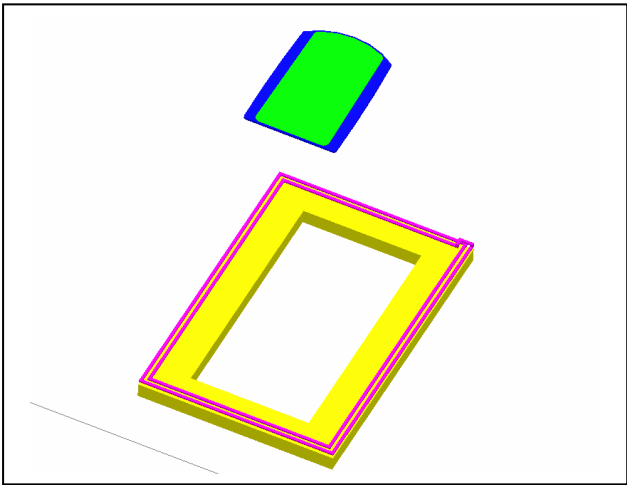


Figure 3. Example of a 3-D FEM model for the calculation of mutual inductance

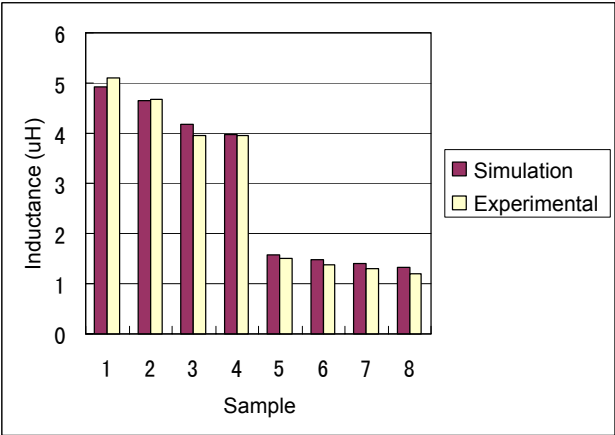


Figure 1. Comparison between simulations and experiments on inductance

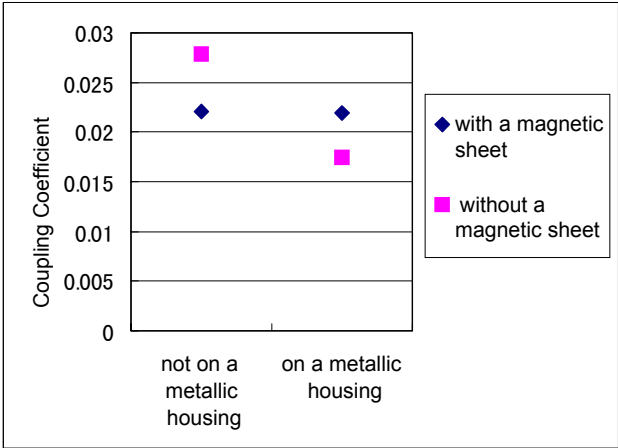


Figure 4. The effect of the magnetic sheet

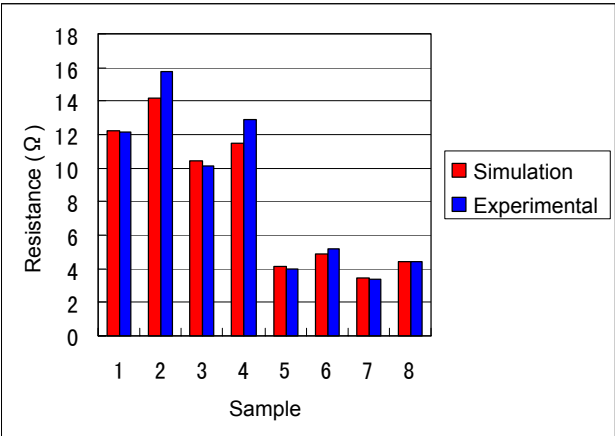


Figure 2. Comparison between simulations and experiments on resistance

III. DESIGN OF RFID DEVICES

A. *Basic structure of the reader/writer*

The electromagnetic transformer coupling between the reader and the card operates the card reader. So, the primary goal in antenna design is high power transfer efficiency. To achieve this, we focused on the minimizing of eddy-current loss. To minimizing eddy-current loss in a metallic housing, we positioned the magnetic sheet between the antenna and the metallic housing. The magnetic sheet is under the antenna coil loop and is a guide for the magnetic flux generated by the coil. Therefore, most of the magnetic flux does not go into the metallic housing. However, Null points occur in the power-supply transmission curves.

B. Structure of new reader

To minimize the Null points, the magnetic sheet is positioned across the antenna coil, as is shown in Figs. 5 and 6. The magnetic sheet placed across the antenna coil acts as the magnetic core of the coil. From operating side, we can't see the loop of the coil, so there is no Null point of power transfer. Fig. 7 shows the magnetic field of this design.

This structure is relatively easy to produce. To make this device as thin as possible, we developed a new magnetic material that has high real permeability and low imaginary permeability. The target specifications for the magnetic material were based on the FEM simulation results. A PDA can only read a card if the card is in a specified position. To make it easier we optimized the line/space of the reader coil. Figure 8 shows the difference between the normalized coupling coefficient of the conventional design and of the new design. The horizontal axis shows the center offset between the reader/writer and tag. The magnetic material has another function; it is a shield from and for internal electrical circuit.

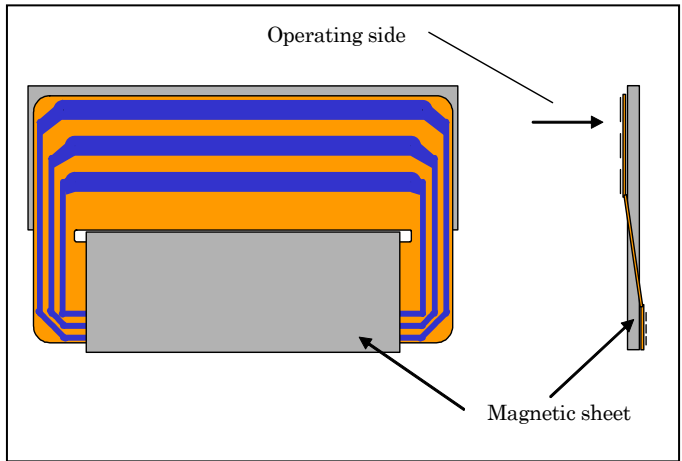


Figure 5. Structure of Reader/Writer

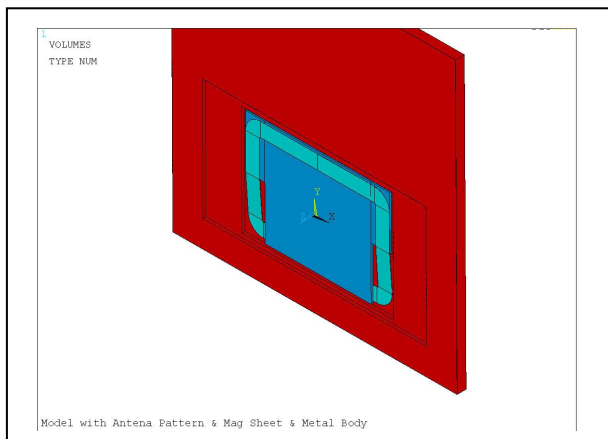


Figure 6. FEM model of reader/writer

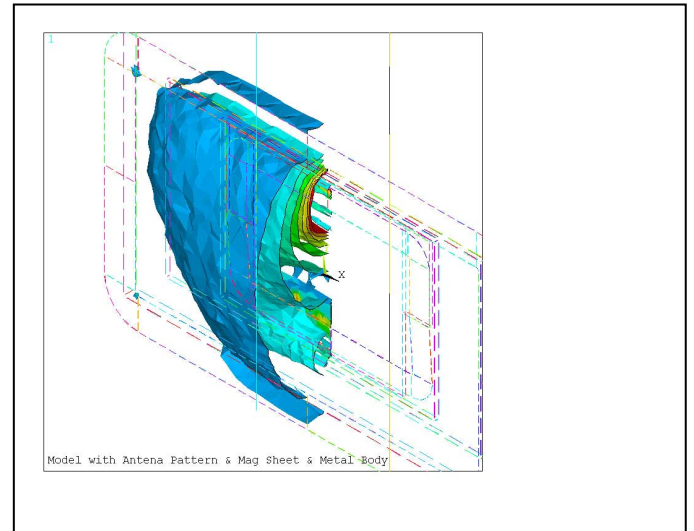


Figure 7. The magnetic field around the reader/writer

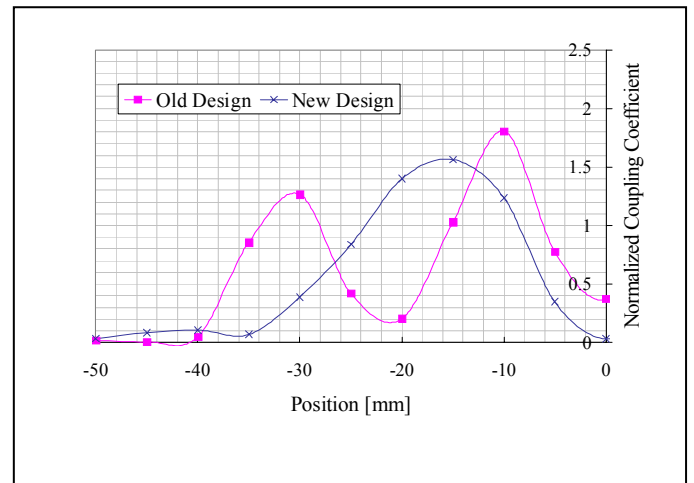


Figure 8. Comparison of the coupling coefficient of the old and new design

IV. RFID DEVICE DRIVERS AND NETWORK-LIBRARY SOFTWARE

Using an RFID device and the new software, a client is connected to the server of a prepaid card via the Internet. Secure identification of the user and secure transmission between the server and the client enables users to buy goods using electronic money. We have developed network-Library and reader/writer device driver software as shown in Fig. 9. The network-library controls access to a high security server and supports HTTP communication. When HTTP communication is carried out, encryption and authentication are performed using SSL (Secure Sockets Layer). The library software can operate with a variety of systems, such as a prepaid-card or credit card.

From the implementation point of view, the driver software has a three-layered structure as shown in Fig. 10. The interface layer makes it easy to install programs in various operation systems. Core software controls devices connected to IC cards or reader/writers, including the setup between devices. The plug-in layer is designed so that driver software can easily connect to a variety of network-library such as a pre-paid card or credit card.

These two new software programs take advantage of the availability of AV/IT mobile systems such as PDAs and mobile phones with low CPU power.

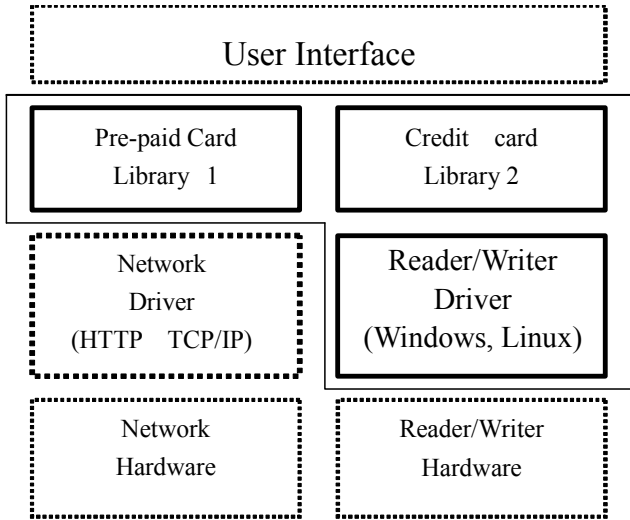


Figure 9. RFID Software

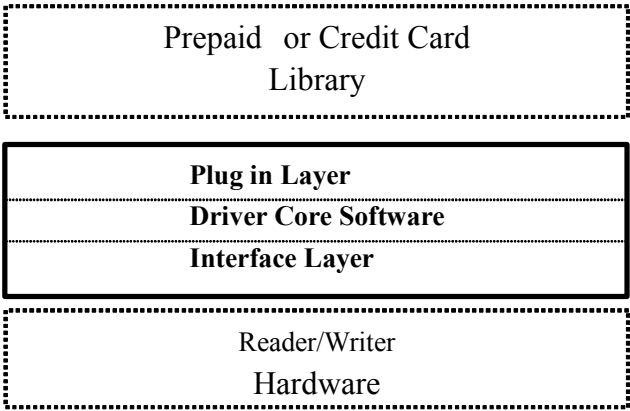


Figure 10. Driver software architecture

V. CONCLUSION

Using electromagnetic field simulation based on FEM, we developed a new small, thin RFID device. The proposed key software with layer architecture can be applied to mobile systems. The device and the software provide secure identification of the user and secure transmission between the server and mobile systems.

ACKNOWLEDGMENT

This work required the cooperation of the staff of RMC and FBC, which we are grateful for.

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